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What is the docker engine version? *(4 points)*

20.10.7

Correct! ✔

```
ubuntu@ip-10-0-1-183:~$ docker --version
Docker version 20.10.7, build 20.10.7-0ubuntu5~18.04.3
```

What is ID of the container using an alpine image? *(4 points)*

14a40ad9fc97

Correct! ✔

```
ubuntu@ip-10-0-1-183:~$ sudo docker ps -a
CONTAINER ID   IMAGE                      COMMAND                  CREATED        STATUS
               PORTS                                               NAMES
2a681f29b74f   mysql                      "docker-entrypoint.s…"   4 years ago    Restarting (1) 31
 seconds ago                                                       ducker_db_1
7bf791519721   adminer                    "entrypoint.sh docke…"   4 years ago    Up 57 minutes
               0.0.0.0:8080->8080/tcp, :::8080->8080/tcp           ducker_adminer_1
396ead30f59d   photon                     "/bin/bash"              4 years ago    Exited (0) 4 year
s ago                                                              not_electron
c5932d260aa9   logstash:7.9.0            "/usr/local/bin/dock…"   4 years ago    Exited (255) 3 ye
ars ago        5044/tcp, 9600/tcp, 0.0.0.0:9885->5000/tcp          pedantic_lumiere
202d3491d369   redis                      "docker-entrypoint.s…"   4 years ago    Exited (255) 3 ye
ars ago        6379/tcp                                            busy_feynman
14a40ad9fc97   python:3.7.10-alpine3.12   "python3"                4 years ago    Exited (0) 4 year
s ago                                                              clever_murdock
ubuntu@ip-10-0-1-183:~$ ^C
```

What networks use bridge driver? *(4 points)*

bridge,devenet, ducker_default

Correct! ✔

```
ubuntu@ip-10-0-8-95:~$ sudo docker network ls
NETWORK ID     NAME              DRIVER    SCOPE
cee1a78284f8   bridge            bridge    local
e2081d082595   devnet            bridge    local
ea06a0dd1728   ducker_default    bridge    local
f404811e0127   host              host      local
2fdf3a7eefa1   none              null      local
ubuntu@ip-10-0-8-95:~$ ^C
```

Which network is associated with the redis container? Mention Gateway IP as well. *(4 points)*

devnet, 172.18.0.2

Correct! ✔

```
ubuntu@ip-10-0-8-95:~$ sudo docker ps -a
CONTAINER ID   IMAGE                       COMMAND                  CREATED
 STATUS                          PORTS                                          NAM
ES
2a681f29b74f   mysql                       "docker-entrypoint.s…"   4 years ago
 Restarting (1) 5 seconds ago                                                    duc
ker_db_1
7bf791519721   adminer                     "entrypoint.sh docke…"   4 years ago
 Up 34 minutes                   0.0.0.0:8080->8080/tcp, :::8080->8080/tcp      duc
ker_adminer_1
396ead30f59d   photon                      "/bin/bash"              4 years ago
 Exited (0) 4 years ago                                                          not
_electron
c5932d260aa9   logstash:7.9.0              "/usr/local/bin/dock…"   4 years ago
 Exited (255) 3 years ago        5044/tcp, 9600/tcp, 0.0.0.0:9885->5000/tcp     ped
antic_lumiere
202d3491d369   redis                       "docker-entrypoint.s…"   4 years ago
 Exited (255) 3 years ago        6379/tcp                                        bus
y_feynman
14a40ad9fc97   python:3.7.10-alpine3.12    "python3"                4 years ago
 Exited (0) 4 years ago                                                          cle
ver_murdock
```

```
ubuntu@ip-10-0-8-95:~$ sudo docker inspect 202d3491d369
```


```
            "Networks": {
                "devnet": {
                    "IPAMConfig": null,
                    "Links": null,
                    "Aliases": [
                        "202d3491d369"
                    ],
                    "NetworkID": "e2081d082595bc736dff38d3545178af7a15a28437e998
a75ce862840e765ca7",
                    "EndpointID": "28df5a5157cd2066f04b39a86265e1262e2a5cffeafe5
e49d80933dc4718911d",
                    "Gateway": "172.18.0.1",
                    "IPAddress": "172.18.0.2",
                    "IPPrefixLen": 16,
                    "IPv6Gateway": "",
                    "GlobalIPv6Address": "",
                    "GlobalIPv6PrefixLen": 0,
                    "MacAddress": "02:42:ac:12:00:02",
                    "DriverOpts": null
                }
```


What host and container ports are mapped in the pedantic_lumiere container? *(4 points)*

9885, 5000

Correct! ✔

```
ubuntu@ip-10-0-8-95:~$ sudo docker ps -a
CONTAINER ID   IMAGE                      COMMAND                  CREATED       STATUS                          PORTS
                                NAMES
2a681f29b74f   mysql                      "docker-entrypoint.s…"   4 years ago   Restarting (1) 4 seconds ago
                                ducker_db_1
7bf791519721   adminer                    "entrypoint.sh docke…"   4 years ago   Up 49 minutes                   0.0.0.0:8080->80
80/tcp, :::8080->8080/tcp       ducker_adminer_1
396ead30f59d   photon                     "/bin/bash"              4 years ago   Exited (0) 4 years ago
                                not_electron
c5932d260aa9   logstash:7.9.0             "/usr/local/bin/dock…"   4 years ago   Exited (255) 3 years ago        5044/tcp, 9600/t
cp, 0.0.0.0:9885->5000/tcp      pedantic_lumiere
202d3491d369   redis                      "docker-entrypoint.s…"   4 years ago   Exited (255) 3 years ago        6379/tcp
                                busy_feynman
14a40ad9fc97   python:3.7.10-alpine3.12   "python3"                4 years ago   Exited (0) 4 years ago
                                clever_murdock
ubuntu@ip-10-0-8-95:~$ sudo docker inspect ^C
ubuntu@ip-10-0-8-95:~$ sudo docker inspect c5932d260aa9
[
    {
        "Id": "c5932d260aa9098d14a6e7be0f22659dda84efecebfdc139ece7cb38851b83ee",
        "Created": "2021-06-11T08:42:04.700352236Z",
        "Path": "/usr/local/bin/docker-entrypoint",
        "Args": [],
        "State": {
            "Status": "exited",
            "Running": false,
            "Paused": false,
            "Restarting": false,
```

```
            "LinkLocalIPv6PrefixLen": 0,
            "Ports": {
                "5000/tcp": [
                    {
                        "HostIp": "0.0.0.0",
                        "HostPort": "9885"
                    }
                ],
                "5044/tcp": null,
                "9600/tcp": null
            },
```

What is the command executed at startup when the 'Dockerfile-wordpress' is run using docker compose? *(5 points)*

data.sh, check

Correct! ✔

in the Dockerfile, Entrypoint is the main process run when the container is run. CMD provide the default arguments for Enterpoint.

```
        chown -R www-data:www-data /usr/src/wordpress
COPY data.sh /usr/local/bin/
ENTRYPOINT ["data.sh"]
CMD ["check"]
```

What is the database password for the container running Dockerfile-mysql.yml? *(5 points)*

AreY0uLookIng4m3

Correct! ✔

```
ubuntu@ip-10-0-8-95:~/Desktop/Ducker$ cd secrets
ubuntu@ip-10-0-8-95:~/Desktop/Ducker/secrets$ ls -la
total 16
drwxr-xr-x 3 ubuntu ubuntu 4096 Jun 11  2021 .
drwxr-xr-x 3 ubuntu ubuntu 4096 Jun 11  2021 ..
---------- 1 root   root    122 Jun 11  2021 .env
drwxr-xr-x 2 root   root   4096 Jun 11  2021 html
ubuntu@ip-10-0-8-95:~/Desktop/Ducker/secrets$ cat .env
cat: .env: Permission denied
ubuntu@ip-10-0-8-95:~/Desktop/Ducker/secrets$ sudo cat .env
DB_CONNECTION=mysql
DB_HOST=127.0.0.1
DB_PORT=3306
DB_DATABASE=krusty
DB_USERNAME=krab
DB_PASSWORD=AreY0uLook1ng4m3
```

What command would you execute to limit the number of active processes to 58 inside a container? *(5 points)*

--pid-limit 58

Correct! ✔

Execute the `docker run` command with the `--pids-limit` flag. This flag sets a hard limit on the total number of processes and threads allowed within the container's control group (cgroup). Docker Docs +2

- **Command:**
  `docker run --pids-limit 58 <image_name>`
- **Verification:**
  You can verify the active count by running the docker container stats command and checking the **PIDS** column. Docker Docs +1

How do you prevent processes from gaining new privileges in the container? *(5 points)*

--security-opt no-new-privileges

Correct! ✔

To prevent processes from gaining new privileges in a Docker container, the primary method is using the `no-new-privileges` security option. This flag ensures that a process and its children cannot escalate their privileges beyond what they had at the start, effectively blocking `setuid` and `setgid` binaries from working. GitHub +2

Primary Prevention Methods

- **Use the `--security-opt` Flag**: When starting a container via the CLI, include `--security-opt no-new-privileges`.

```bash
docker run --security-opt no-new-privileges <image-name>
```

How do you pull mysql image with hashsum "sha256:68b207d01891915410db3b5bc1f69963e3dc8f23813fc ? *(5 points)*

docker pull mysql@sha256:sha256:68b207d Correct! ✔

2. Pull the Image by Digest

Replace `<hashsum>` with the actual SHA256 string (e.g., `sha256:548da4c67...`).

```bash
docker pull mysql@sha256:<hashsum>
```

Where are the logs for the container 'not_electron' located? *(5 points)*

/var/lib/docker/containers/396ead30f59d88 Correct! ✔

```
ubuntu@ip-10-0-8-95:~$ sudo docker ps -a
CONTAINER ID   IMAGE                       COMMAND                  CREATED       STATUS                          PORTS
                                NAMES
2a681f29b74f   mysql                       "docker-entrypoint.s…"   4 years ago   Restarting (1) 44 seconds ago
                                ducker_db_1
7bf791519721   adminer                     "entrypoint.sh docke…"   4 years ago   Up 3 hours                      0.0.0.0:8080->8
080/tcp, :::8080->8080/tcp    ducker_adminer_1
396ead30f59d   photon                      "/bin/bash"              4 years ago   Exited (0) 4 years ago
                                not_electron
c5932d260aa9   logstash:7.9.0              "/usr/local/bin/dock…"   4 years ago   Exited (255) 3 years ago        5044/tcp, 9600/
tcp, 0.0.0.0:9885->5000/tcp   pedantic_lumiere
202d3491d369   redis                       "docker-entrypoint.s…"   4 years ago   Exited (255) 3 years ago        6379/tcp
                                busy_feynman
14a40ad9fc97   python:3.7.10-alpine3.12    "python3"                4 years ago   Exited (0) 4 years ago
                                clever_murdock
ubuntu@ip-10-0-8-95:~$ sudo docker inspect 396ead30f59d
[
    {
        "Id": "396ead30f59d887c3434758858972cde07b151f24151383c1d6f4040969c8e2a",
        "Created": "2021-06-11T08:42:37.21916417Z",
        "Path": "/bin/bash",
        "Args": [],
```

```
        "HostnamePath": "/var/lib/docker/containers/396ead30f59d887c3434758858972cde07b151f24151383c1d6f4040969c8e2a/hostname",
        "HostsPath": "/var/lib/docker/containers/396ead30f59d887c3434758858972cde07b151f24151383c1d6f4040969c8e2a/hosts",
        "LogPath": "/var/lib/docker/containers/396ead30f59d887c3434758858972cde07b151f24151383c1d6f4040969c8e2a/396ead30f59d887c
3434758858972cde07b151f24151383c1d6f4040969c8e2a-json.log",
        "Name": "/not_electron",
```